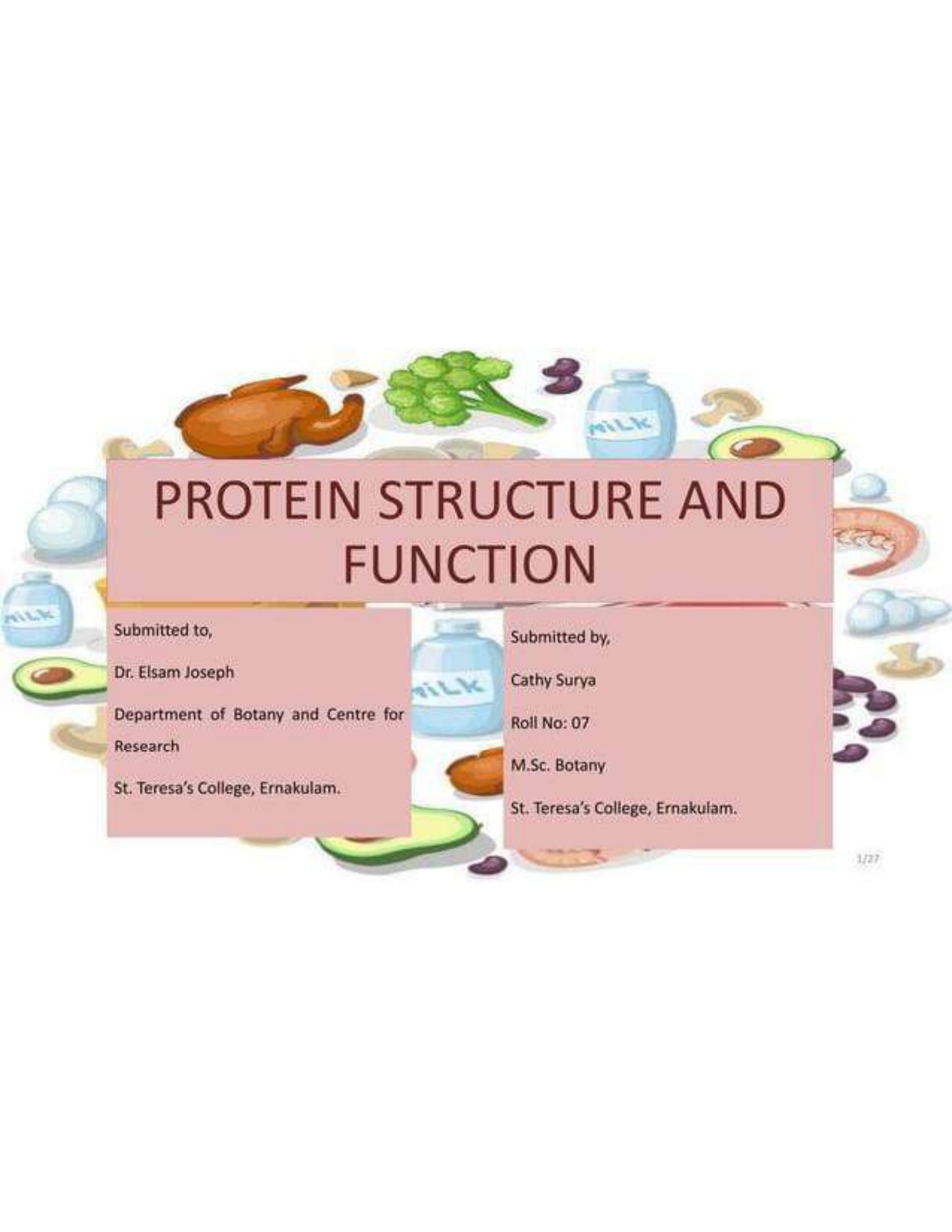




PROTEIN STRUCTURE AND FUNCTION

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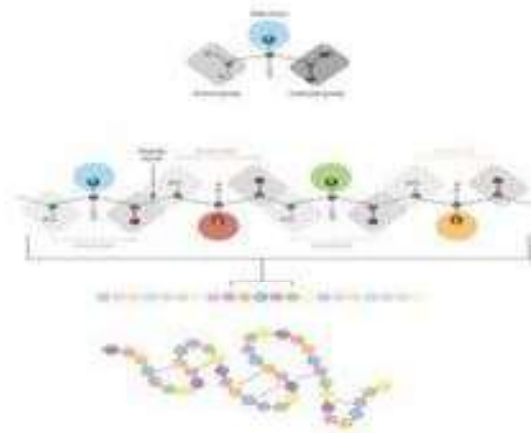


Proteins

- Proteins are most abundant structural and functional organic constituents of living matter.
- They constitute the fundamental basis of structure and function of life.
- Proteins are the end products of the decoding process that starts with the information in cellular DNA.
- **G. J Mulder** – First to describe about proteins.

What Are Proteins Made Of?

- Proteins are built of monomer units called **amino acids**, which are small organic molecules that consist of an α (central) C atom linked to an NH_3 group, a carboxyl group, a H_2 atom, and a variable component called a side chain (R).
- Proteins are formed of C, H, O and N with S and P in some cases.
- 300 kinds of amino acids are known to exist in nature. Of these, 20 kinds involves in the formation of proteins called **proteogenic amino acids**.

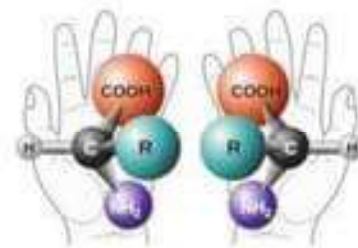
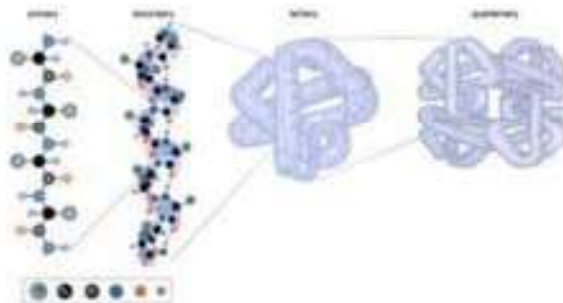


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- These are all α -amino acids (only differ in R group) and are chiral.
- The two mirror-image forms are called the l isomer and the d isomer.
- Only l amino acids are constituents of proteins.

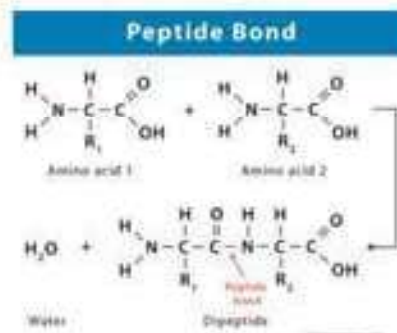
Biochemists have distinguished several level of structural organization of proteins.

1. Primary structure.
2. Secondary structure
3. Tertiary structure
4. Quaternary structure



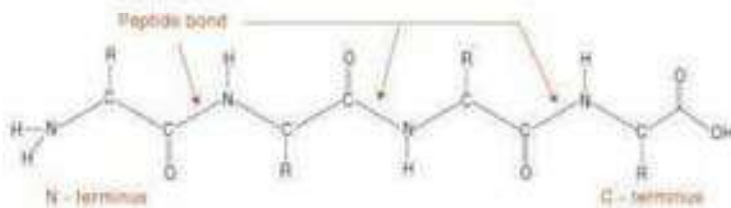
Peptide bond

- The peptide bond is essentially planar- The six atoms in amino acids lie in the same plane: the α -carbon atom and CO group from the first amino acid and the NH group and α -carbon atom from the second amino acid.
- Peptide bond formation – It is accompanied by the loss of a molecule of water.
- Peptide bonding is an endergonic process.



Primary Structure

- Sequence of amino acids present in the polypeptide chain.
- Determined by covalent bond, protein is linear.
- The primary structure of protein starts from amino acid terminal (N) end and ends in the carboxyl group (C) end.
- Linear polymer of amino acids are polar.
- Each component amino acid in a polypeptide is called a **residue**.
- Repeating unit of polypeptide chain is known as back bone.
- Variable portion of polypeptide chain are the side chains.



Primary Structure (Cont...)

- Polypeptide has the ability to form many hydrogen bonds.

Hydrogen bond doner: N-H group
Hydrogen bond acceptor: C=O group

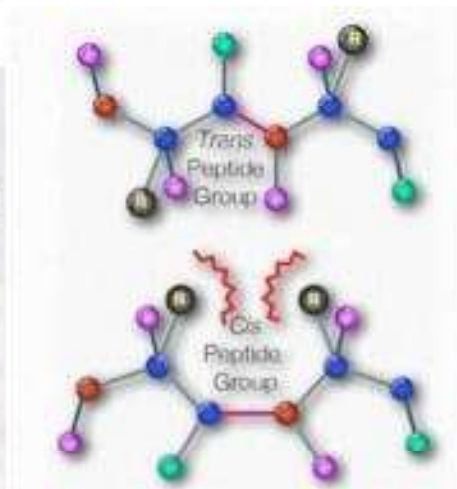
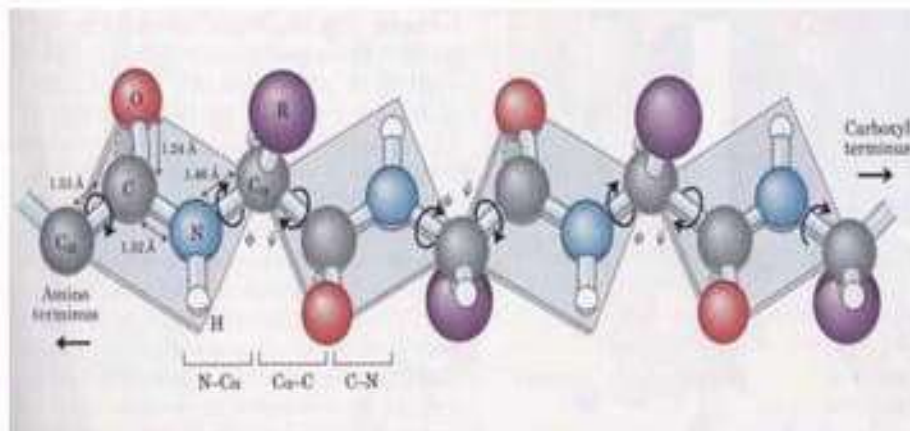
- Peptide bonds are resonance stabilized (have double bond character)

This double bond nature of peptide bond:

- Makes the peptide bond planar.
- Prevents rotation on the peptide bond

Continuous..

- Almost all peptide bonds in proteins are trans.(because of the steric hinderance occurring on the cis structure).

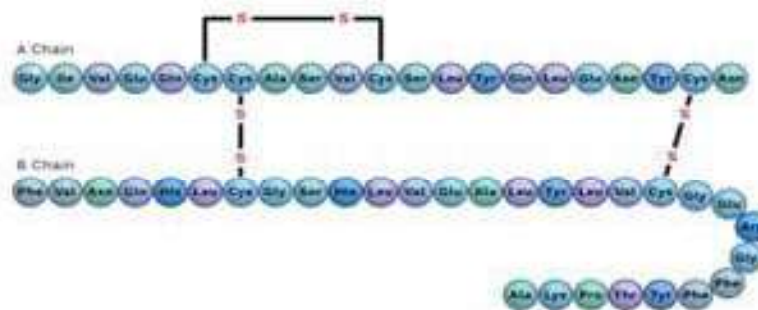


Functions

- The amino acid sequence within the polypeptide chain is crucial for the protein's proper functioning.
- This sequence is encrypted in the DNA genetic code.
- If mutation is present in the DNA and the amino acid sequence is changed, the protein function may be affected.
- All documented genetic disorders, such as cystic fibrosis, sickle cell anemia, albinism, etc., are caused by mutations resulting in alterations in the primary protein structures, which in turn lead to alterations in the secondary , tertiary and probably quaternary structure.

Frederick Sanger

- He said that the two polypeptide chains of the protein insulin had precise amino acid sequences and, by extension, that every protein had a unique sequence.



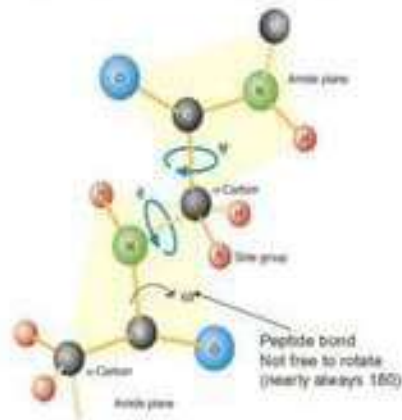
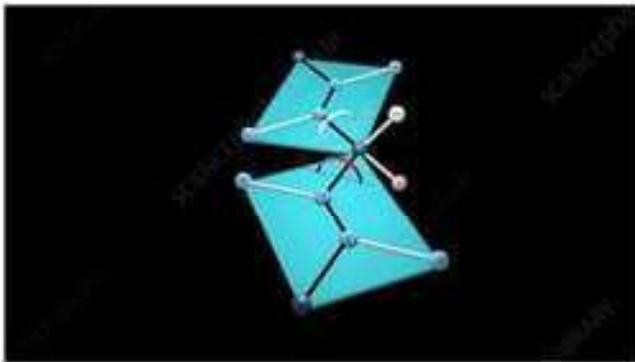
Secondary Structure

- The primary structure of protein twists into regular patterns forming secondary structure.
- These twists are formed due to the formation of hydrogen bond between N=H and C=O groups on the polypeptide chain.
- These are found in – Keratin of hair, Collagen, elastin
- The structures include
 - α helix
 - β sheet
 - β pleated sheets
 - β bends
- Linus Pauling and Robert Corey proposed two periodic structures called the α helix (alpha helix) and the β pleated sheet.



Polypeptide chain conformations

- The reasonable free movements are the rotations around $C\alpha-N$ bond (ϕ) and $C\alpha-C$ bond (ψ).
- The conformation of the backbone can be described by torsional angles/dihedral angles.



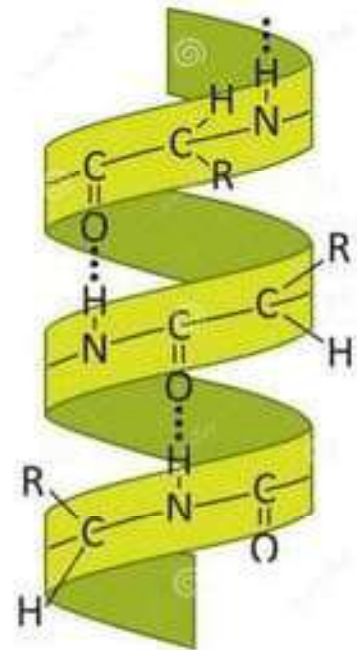
Alpha helix

- Right handed spiral structure.
- It is rodlike, cylindrical.

Tightly coiled backbone - Inner part of the rod.

Side chains - Extend outward in a helical array.

- Stabilized by hydrogen bonds between the NH and CO groups of the main chain.
- Hydrogen bonds are almost parallel to the axis of helix.

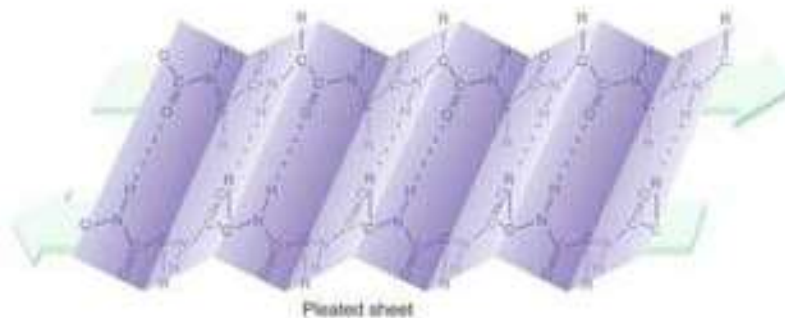


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- The **screw sense** of the α helix describes the direction in which the helix rotates with respect to its axis.
- The right handed α helix predominates because there is less steric hinderance between the side chains.
- Amino acids per turn – 3.6
- Pitch is 5.4 Å
- Distance between adjacent amino acids – 1.5Å
- Alpha helical segments are found in many globular proteins like myoglobin, troponin-C etc.

Beta pleated sheet

- Two or more polypeptide line up side by side.
- Individual polypeptide- beta strand.
- Each beta strand is fully extended.
- The distance between adjacent amino acids along a β strand is approximately 3.5 Å.



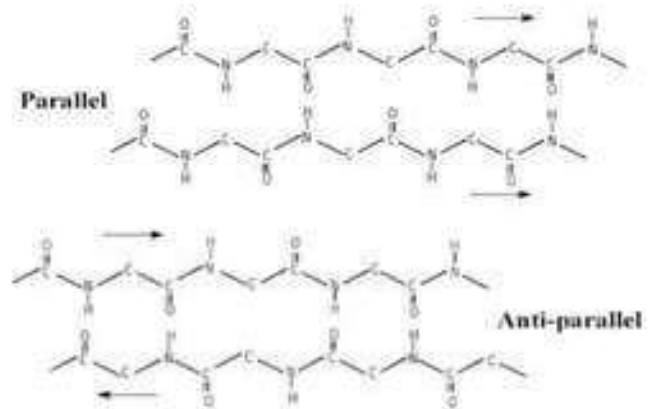
Types of beta pleated sheet

Anti parallel

- Two linear polymer of amino acid run in opposite direction. The NH group and the CO group of each amino acid are respectively hydrogen bonded to the CO group and the NH group of a partner on the adjacent chain

Parallel

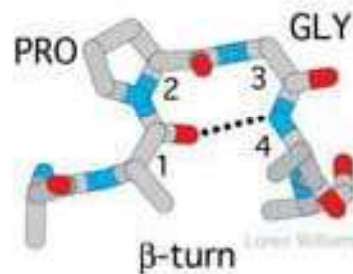
- The adjacent strands run in the same direction. For each amino acid, the NH group is hydrogen bonded to the CO group of one amino acid on the adjacent strand, whereas the CO group is hydrogen bonded to the NH group on the amino acid two residues farther along the chain



Fatty acid-binding proteins, important for lipid metabolism, are built almost entirely from β sheets.

Beta turn

- Permits change of direction of peptide chain to get a folded structure.
- It gives a protein globularity rather than linearity.
- Hydrogen bond stabilizes the beta bend structure.
- Proline and Glycine are frequently found in beta turns.
- Beta turns often promote the formation of antiparallel beta sheets.
- Occur at protein surfaces.



Tertiary Structure

- It is the 3D conformation of a polypeptide.
- The structure is based on various types of interactions between the side chains of the peptide chain.

Two classes of protein based on Tertiary structure

Globular Protein

- Compact, spherical or globular shape.
- Soluble in water.
- The structure is unstable.
- Most proteins within the cell are globular.
- Examples: Enzymes, Myoglobin.



Fibrous Protein

- Elongated
- Insoluble in water.
- The structure is stable.
- Most proteins outside living cells are fibrous proteins.
- Examples: Collagen, Elastin



Fibrous proteins

Tertiary structure

- Most stable and found in most biologically active proteins.
- It is important in that biological properties such as enzyme activity, antigenicity etc.

Myoglobin: The First Globular Protein Whose Tertiary Structure Was Determined

- The detailed tertiary structure of a protein is usually determined using the technique of X-ray crystallography.
- Myoglobin was the first protein to be seen in atomic detail.
- Myoglobin, the oxygen carrier in muscle, is a single polypeptide chain of 153 amino acids.
- Myo-globin is an extremely compact molecule.

Myoglobin: The First Globular Protein Whose Tertiary Structure Was Determined

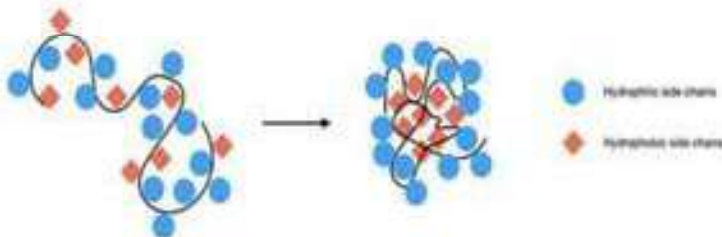
- About 70% of the main chain is folded into eight α helices, and much of the rest of the chain forms turns and loops between helices.
- The interior consists almost entirely of nonpolar residues such as leucine, valine, methionine, and phenylalanine
- The only polar residues inside are two histidine residues, which play critical roles in binding iron and oxygen.
- The outside of myoglobin, consists of both polar and nonpolar residues.

Myoglobin

The process of folding proteins into their tertiary structures is spontaneous and involves bonds and intermolecular forces to make the structure stable.

Hydrophobic interactions

- Most proteins exist in aqueous solution.
- The amino acids with hydrophobic side chains (valine, leucine etc.) will tend to be found inside the protein.
- The amino acids with hydrophilic side chains (lysine, aspartate etc.) will tend to be found outside the protein.



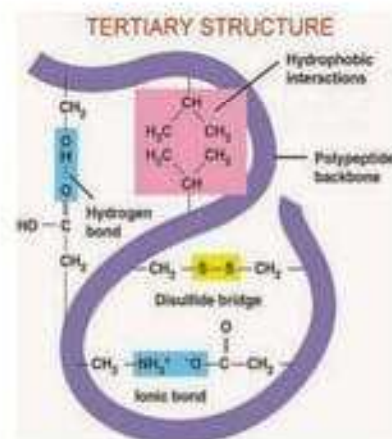
A protein in an aqueous environment will have a hydrophobic core and a hydrophilic surface.

Continuous...

- **Van der Waals interactions** between tightly packed hydrocarbon side chains also contribute to the stability of proteins.

Disulfide bonds

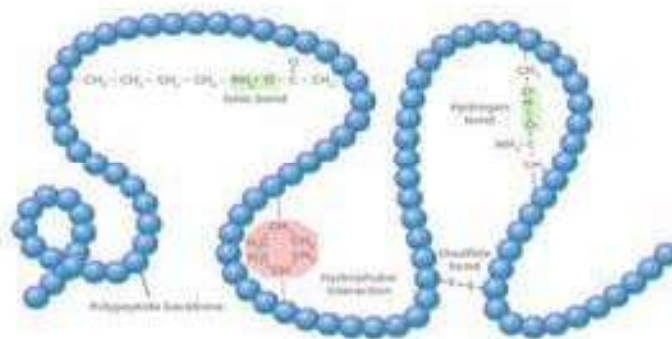
- Disulfide bonds are formed between two sulfur (SH) atoms, which are found in the side-chain of the amino acid cysteine.
- When two cysteines are brought into close proximity in the tertiary structure, an enzyme called “protein disulfide isomerase” forms a disulfide bond between the two SH groups.
- These cross-linked units are called *cystines*.



Continuous...

Hydrogen bonds

- Atoms, such as oxygen, can be covalently bound to hydrogen, and giving the hydrogen (H) in the OH group a slight positive charge due to the oxygen (O) attracting electrons.
- When this OH group comes across another atom with a slightly negative charge, such as oxygen, the hydrogen from the OH group reaches out to the oxygen atom and forms a bond.
- This is known as a hydrogen bond, and occurs between amino acids which have what is called a polar side-chain.



Quaternary Structure

- The quaternary structure is usually determined by X-ray crystallography.
- The quaternary structure of a protein is the association of several protein chains or subunits into a closely packed arrangement.
- Each of the subunits has its own primary, secondary, and tertiary structure.
- The subunits are held together by hydrogen bonds and van der Waals forces between nonpolar side chains.
- The simplest sort of quaternary structure - **Dimer**, consisting of two identical subunits.



Figure 3.48. Quaternary Structure. The Cro protein of bacteriophage λ is a dimer of identical subunits.

Quaternary Structure

- More than one type of subunit can be present, often in variable numbers.

Example, **Human hemoglobin**, consists of two subunits of one type (designated α) and two subunits of another type (designated β).

The coat of rhinovirus, the virus that causes the common cold, includes 60 copies each of four subunits.

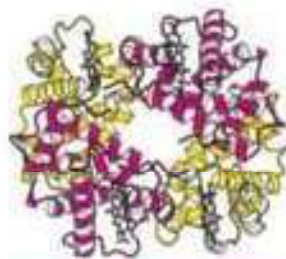
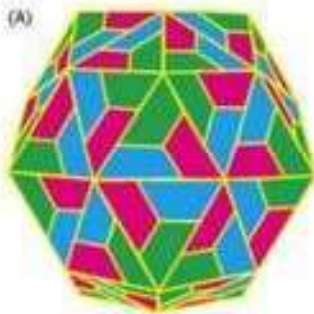


Figure 3.45. The $\alpha_2\beta_2$ Tetramer of Human Hemoglobin. The structure of the two identical α subunits (red) is similar to but not identical with that of the two identical β subunits (yellow). The molecule contains four heme groups (black with the iron atom shown in purple).

References

- Berg M.G, Tymoczko L.J, Stryer L, *Biochemistry*(5th edition),W.H Freeman and Company (2001).
- David Eisenberg, *The discovery of the α -helix and β -sheet, the principal structural features of proteins* (2003).
- Watson, J., Baker, T., Bell, S., Gann, A., Levine, M., & Losick, R. *Molecular Biology of the Gene* (7th ed.). Boston: Pearson (2007).
- <https://www.nature.com/scitable/topicpage/protein-structure>
- <https://comis.med.uvm.edu/>

THANK YOU